High-Dimensional Visual Embedding for Language-Aligned Perception

Vision Encoder → LLM Decoder for Maze Reasoning

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Goal

Demonstrate that rich visual embeddings may contain enough structural information for an LLM to resolve structured spatial reasoning tasks.

Pipeline Overview:

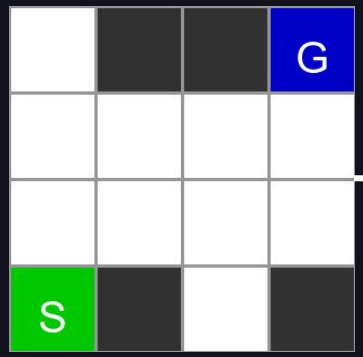
1. **Input:** Raw maze image (.png)
2. **Vision Encoder:** A ResNet-18 network extracts high-dimensional visual embeddings that capture spatial layout, walls, and goal positions.
3. **Embedding Projection:** Embeddings are transformed into **prefix tokens** compatible with a language-model decoder.
4. **LLM Decoder:** A GPT-2 decoder autoregressively generates the corresponding **action sequence**.

Output: A sequence of discrete moves representing the solution path through the maze. [U R R U R R]

Pipeline

[U, U, R, R, R, U]

Generate Mazes



ResNet18 (Encoder)

Image Embedding

GPT2 (Decoder)

Exact Solution

[U, U, R, R, R, U]

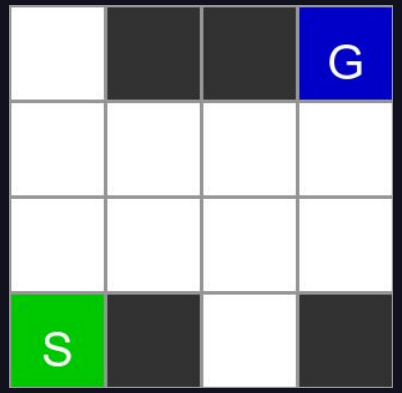
Valid Solution

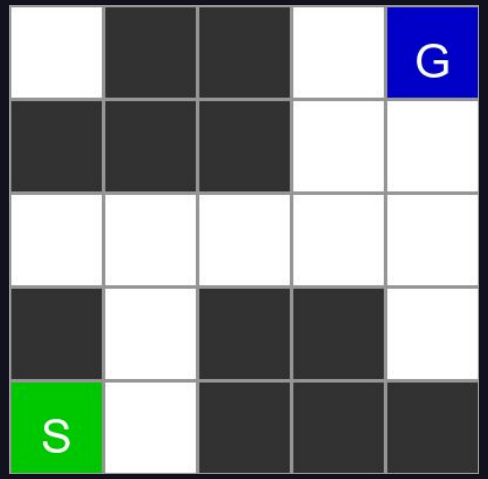
[U, R, R, R, U, U]

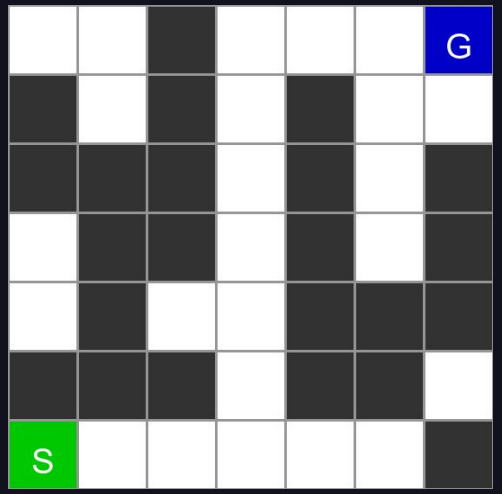
Example: 4x4 Dataset
Total unique solution patterns: 20
Train solution patterns: 16
Test solution patterns: 4
Variations per solution: 150
Total train images: 2400
Total test images: 600

Dataset

- Grid sizes: 4×4 , 5×5, 7×7
- Start: bottom-left ⟶ Goal: top-right
- Actions: R, U only; this limits the number of unique solutions to a maze
  - 4×4: 20 unique sols , 5×5: 70 unique sols, 7×7: 924 unique sols
- Solution Sequence length: 4×4: 6 , 5×5: 8, 7×7: 12







Dataset Generation

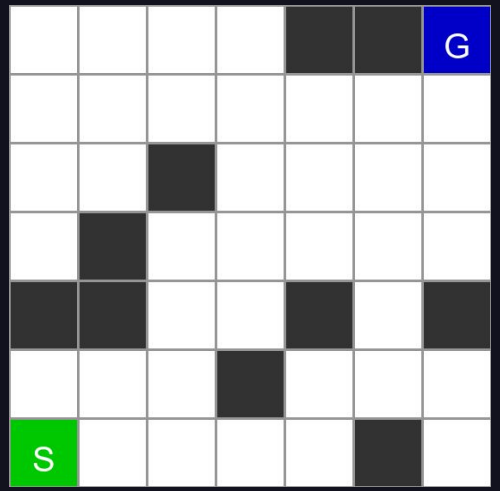
Creating Structurally Different Variations:

- Start with a solution path (e.g., "R R U U R U")
- Walk through path from start (bottom-left) to goal (top-right)
- For each cell NOT on the solution:
  - **Adjacent to path** → 70% chance to block (prevents alternative routes)
  - **Far from path** → 40% chance to block (decorative obstacles)
- Use different random seeds to create 50-150 “unique” maze layouts per solution.
- Result: Same solution, different wall configurations → forces model to learn spatial reasoning, not memorization.

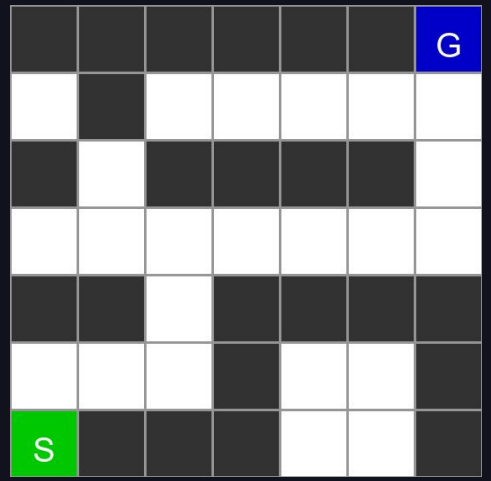
Key: 70% blocking rate on adjacent cells creates just enough variation without making mazes unsolvable

Data Quality Discovery

- Original mazes: random obstacles placements → memorization
- Fix: strategic obstacle placement (strategic placement) → structural diversity
- Model forced to learn real maze-solving

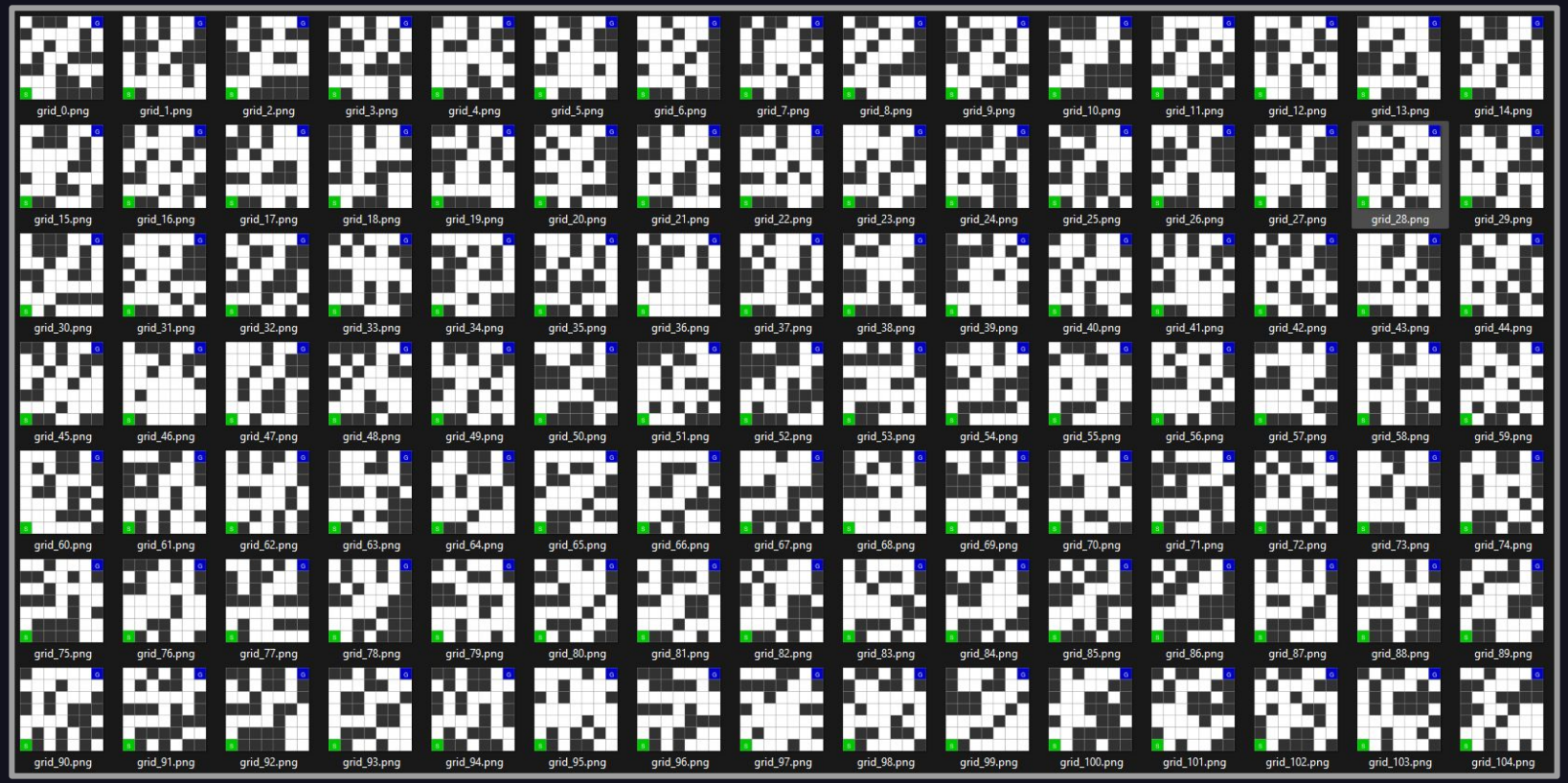


Original



Fix

Grid Variations for dataset diversity



Grids 0 - 49, 50-101, etc.. have the SAME solution, but unique obstacle placement

Do we have duplicates?

4x4 ⟶ 3000 mazes ⟶ 739 mazes with ~2.6 duplicates = 1939
5x5 ⟶ 7000 mazes ⟶ 50 mazes with 2 duplicates = 100
7x7 ⟶ 46200 mazes ⟶ 3 mazes with 2 duplicates = 6

Yes of course we do, but it dramatically decreases because the number of possible maze configurations grows exponentially with grid size.

Model Encoder Selection

- ViT + GPT-2: Domain mismatch (ImageNet → mazes), attention failed to capture path topology → loss plateau 0.34 → 0% accuracy
- ResNet18 + GPT-2: Convolutional inductive bias fits grids; hierarchical features capture paths & walls → better generalization
- Original Experiments:
  - 5×5, 93M → Train 100%, Test 1.9%, (Exact matches)
  - 5×5, 15M → Train 100%, Test 14.8%, (Exact matches)
  - 5×5, 5M → Train 53%, Test 9.6%, (Exact matches)
  - 7×7, 5M → Train 100%, Test 30.9%, (Exact matches)

Architecture Overview

- Vision Encoder: ResNet18 → 512-dim features
- Prefix Generator (MLP): 512 → 128 → 16 soft prompt tokens
- LLM Decoder (GPT2 custom): 2 layers, 2 heads, 128-dim embedding
- GPT-2 Tokenizer Vocabulary: R, U, BOS, EOS, PAD, UNK
  1. **<pad> (ID: 0)** - Padding token to make sequences same length in batches
  2. **<s> (ID: 1)** - BOS (Beginning of Sequence) - signals start of generation
  3. **</s> (ID: 2)** - EOS (End of Sequence) - signals when to stop generating
  4. **<unk> (ID: 3)** - Unknown token (for any unexpected input)
  5. **R (ID: 4)** - Right movement action
  6. **U (ID: 5)** - Up movement action
- Parameters: 11,952,320
- Prefix tokens: 8

Model Optimization / Training

Model Capacity Trials

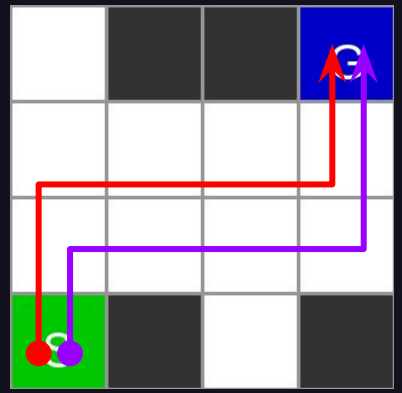
- 93M params → memorized instantly
- 15M params → still memorized
- 11,952,320 params (128-dim) → sweet spot for learning smaller grids (5x5)

Training Setup

- Optimizer: AdamW (ResNet 5e-5, Prefix 5e-4, GPT2 1e-4)
- Cosine annealing scheduler (smooth, performance-optimized LR decay schedule)
- Batch size: 16
- Dropout: 0.4, Gradient clipping: 1.0

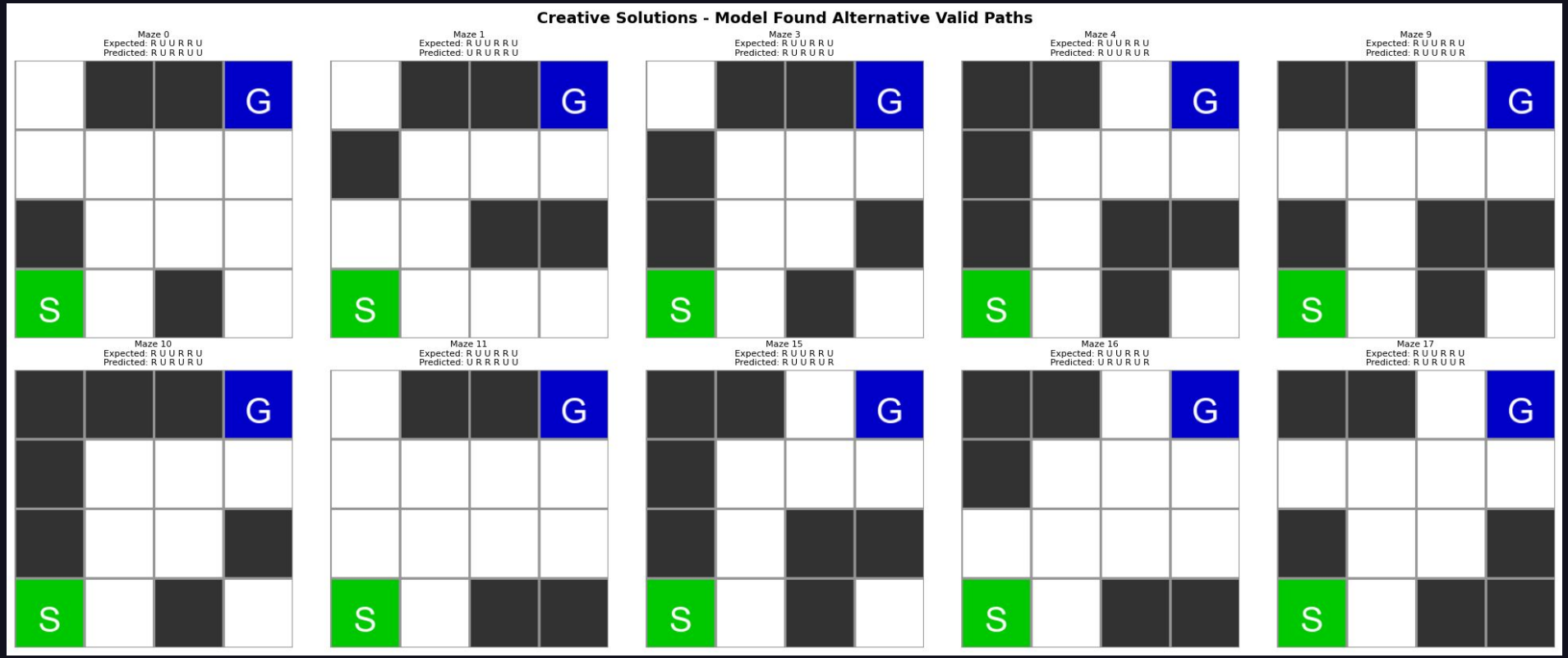
Why Exact Match is Insufficient

- Trivially, multiple valid solutions exist.
- Expected: [U, R, R, R, U, U]
- Predicted: [U, U, R, R, R, U] → different but reaches goal!

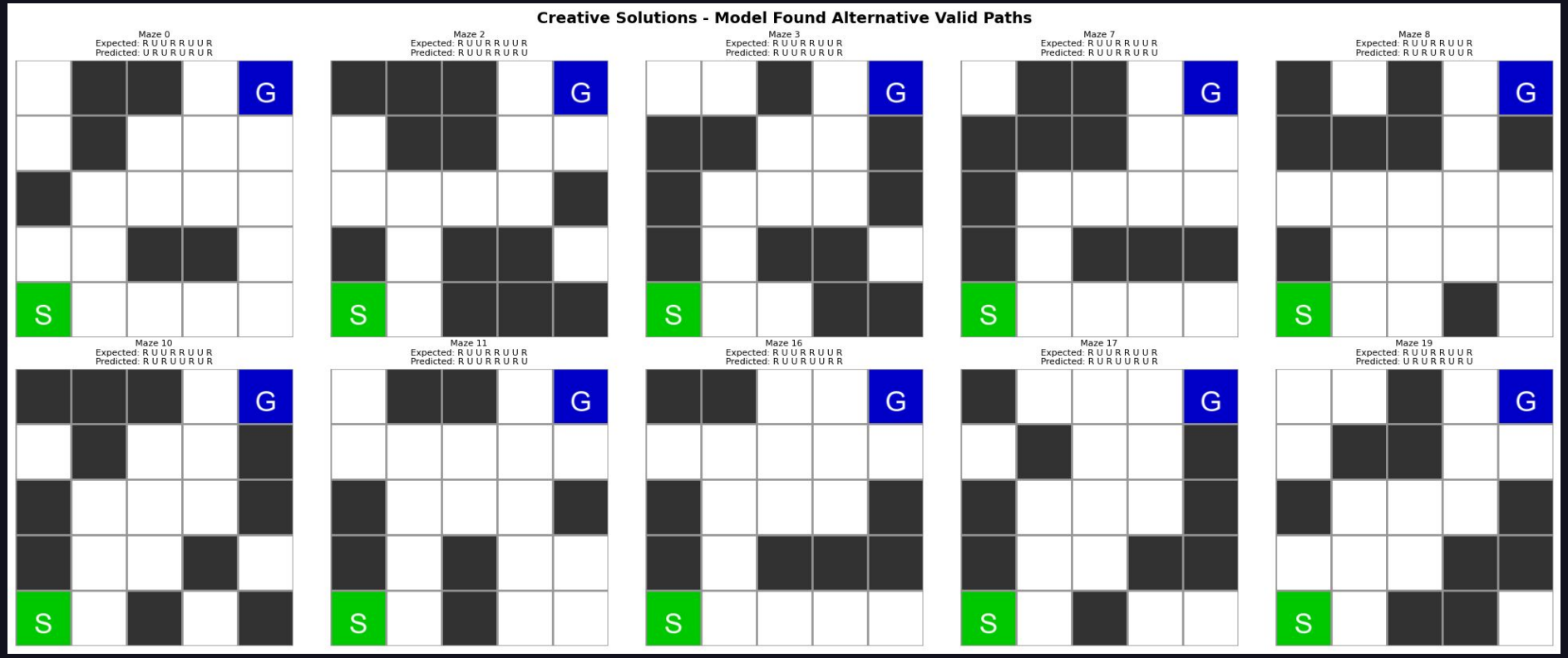


Exact match underestimates model performance by over 50%

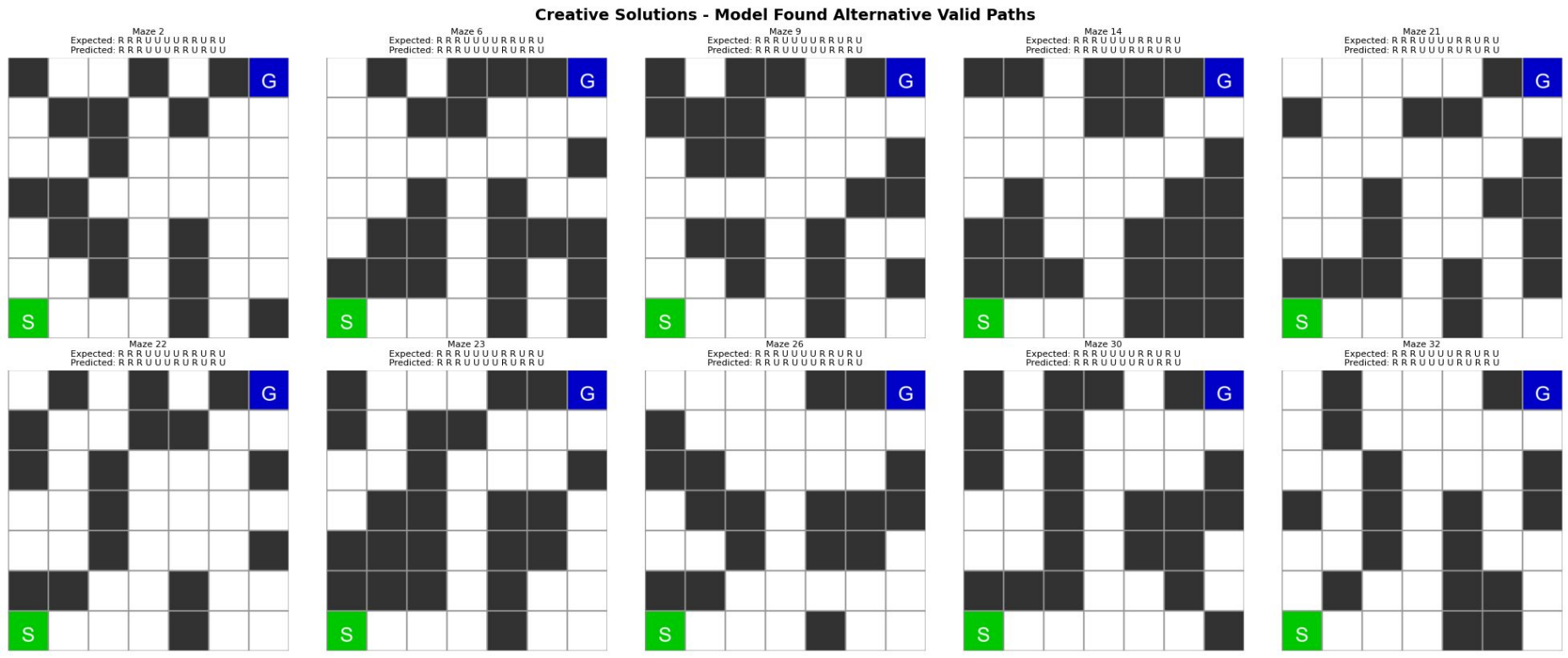
4x4 Creative Solutions



5x5 Creative Solutions



7x7 Creative Solutions



Final Results

Test	Grid	Train Images	Test Images	Model	Epochs	LR	Train Acc	Test Acc (Exact)	Test Acc (Valid)	Gap	Loss	Status
1	4×4	2,400	600	128-dim, 2L, 8 prefix	40	5e-4	71.4%	7.3%	**61.0%**	64.1%	0.1721	⚠️ Moderate
2	5×5	5,600	1,400	128-dim, 2L, 8 prefix	75	5e-4	71.3%	13.1%	**70.1%**	58.2%	0.1459	✅ Good
3	7×7	36,950	9,250	128-dim, 2L, 16 prefix	100	5e-4	23.7%	8.6%	**35.8%**	15.1%	0.2699	❌ Needs Work

Error Analysis

Failure Mode	4×4 Count	5×5 Count	7×7 Count	Total
Hit Wall	230	388	5901	6519
Out of Bounds	4	31	42	77
Invalid Token	0	0	0	0
Didn't Reach Goal	0	0	0	0
Total Failures	234	419	5943	6596

The model's main failure is hitting walls, showing poor spatial encoding, despite correctly learning the action vocabulary, semantics, and sequence length.

Lessons Learned

- Exact matches don’t always showcase model capability.
- Creative solutions prove understanding
- Inductive bias (locality, translation invariance) > pretraining → (ResNet > ViT)
- Structural data diversity is critical
- Having duplicates in small solution space reduces model accuracy. E.g. (4x4)
- Prefix tuning is effective

Future Work

- Push valid solution rate to 90–95%
- Analyze more specific failure paths
- Scale to 10×10, 15×15 grids
- Add Left/Down movements (Increase model complexity)
- Multi-checkpoint mazes
- Attention & embedding visualization
- Apply to robotics / game AI / procedural generation
- Evaluate model performance on visually distinct mazes

Conclusions

- Resnet18 Vision → GPT2 LLM pipeline solves 5x5 mazes with >50% validity
- Our model learns true spatial reasoning
- Validation-based evaluation is essential for multiple-solution tasks